Seaglider Ballasting Procedure

Before a sea trial, a Seaglider needs enough lead trimmed out (or in) that its maximum internal volume (`volmax`) gives the right thrust in the density of the water it's actually deploying into. This page covers the tank procedure for estimating `volmax` and turning it into a weight change. Once the glider is in the water, trim is refined dynamically from flight data — see Trim & Flight Model.

Source

Paraphrased from the APL-UW IOP office-hours session on ballasting and volmax estimation (June 2026) and the APL-UW IOP *SGX Documentation*. The tank method is deliberately simple — IOP's own philosophy is to nail down `volmax` in the tank and let the sea trial work out pitch/roll trim dynamically, rather than trying to model static centers precisely on paper. Numbers quoted here are IOP starting points — confirm against the Parameter Reference Manual and defer to APL-UW IOP guidance.

Equipment

- A freshwater tank large enough to fully submerge the glider **vertically** (deep enough that it can hang without touching bottom or breaking the surface).
- A hanging scale or load cell suspended over the tank, to weigh the glider in water.
- A line to suspend the glider from — tied off at the rudder, since it hangs vertically.
- The glider's comms cable, connected and slack — you need a live link to read/set VBD position while the glider hangs in the tank. Leave the antenna disconnected and simply let it dangle; nothing should be expressing at the surface.

Procedure

1. **Weigh the whole glider dry** — wings, rudder, everything — on a scale in air. This is the mass **M**.